

On the probability distribution in stretched Gaussian like models and exponential tails.

Ahmed Raza Patel*

*13492802@mylife.unisa.ac.za

ABSTRACT

In this paper I will attempt to provide one example of such a power law appearing in the form of a stretched Gaussian distribution or exponential tail in generated data from anagram solutions. Discuss relations to other physical and statistical phenomena and posit whether chaotic and ordered systems produce the same phenomena. We find that for some systems an exponential tail is inevitable, but not for all.

1 Introduction

The asymptotic existence of power laws is a well-established fact in statistical physics and critical phenomena with exact solutions available for the 2D Ising model, for self-avoiding walks, for lattice animals, etc.¹ On the other hand, the relevance of power laws in Nature is less clear-cut even if it has repeatedly been claimed to describe many natural phenomena. In this paper I will attempt to provide one example of such a power law appearing in the form of a stretched Gaussian distribution or exponential tail in generated data from anagram solutions. Discuss relations to other physical and statistical phenomena and posit whether chaotic and ordered systems produce the same phenomena. I will also make reference to similar phenomena in physics for example the abundance of primordial black holes in the early era inflationary models.²

To accomplish showing this relationship let us look at what would happen if we were to plot the relation of solved anagrams generated from a combination of 9 letters of the English alphabet. We will focus on nine letter anagrams, and explain further what exactly solving means in this context later on in the text.

In Figure 1.

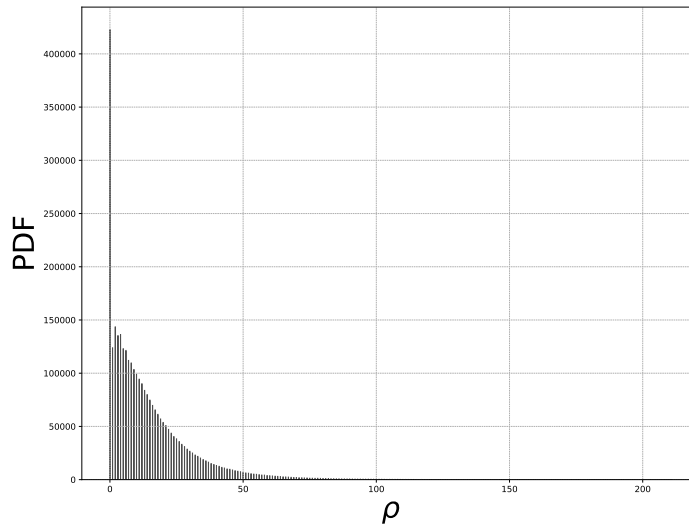


Figure 1. This shows the probability distribution function PDF as a raw output of number of anagrams against the number of the anagrams with solutions ρ here. The graph shows a Gaussian like relation where many anagrams have 0 solutions before exponentially decaying.

2 Evidence of exponential tails in anagram solutions, physics and economy

In Davies³, he shows an exponential tail relation for musicians in the UK top 75, a similar relation to figure 1, if it was plotted on a log log scale this relation would be even clearer, the beginning of the distribution follows a Gaussian like trajectory before decaying exponentially.

Figure 2:

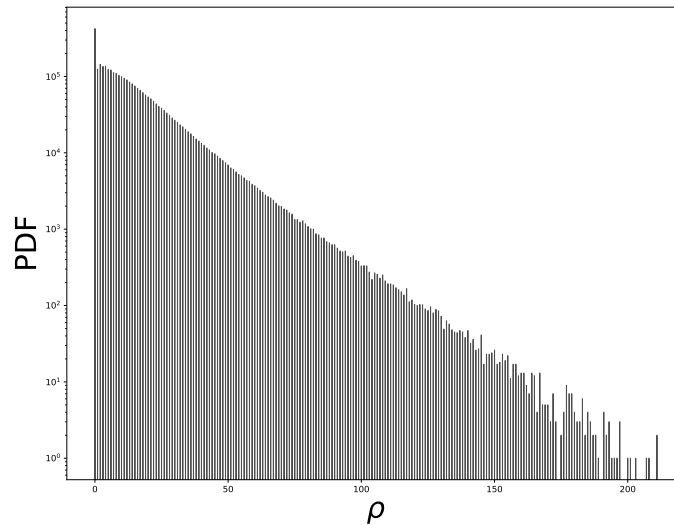


Figure 2. Fig.2 shows the exponential tail as the decay towards the right of the graph. Note by plotting PDF on a log scale one can clearly see how the tail appears almost as a straight line.

Looking at figure 2 above we can clearly see the exponential tail of the data. This happens with many probability distributions as we start off with a high number of solutions for a small set and as the set increases the solutions taper off. Why this happens makes sense as explained in Laherrere⁴ where they show that stretched exponentials describe very well the distributions of radio and light emissions from galaxies, of US GOM OCS oilfield reserve sizes, of World, US and French agglomeration sizes, of country population sizes, of daily Forex US-Mark and Franc-Mark price variations, of Vostok (near the south pole) temperature variations over the last 400 000 years, of the Raup-Sepkoski's kill curve and of citations of the most cited physicists in the world. They explain that Multiplicative processes often constitute zeroth-order descriptions of a large variety of physical systems, exhibiting anomalous pdf's and relaxation behaviors, or in other words the likelihood of a low number of events occurring is higher than the number of many events occurring. This relates to the number of anagrams solvable with a specific letter being required in the solution. As in Melnyk⁵ we can see a similar distribution for entropy shown in figure 1 from the same paper.

3 Solving anagrams using Python

The rules for solving the anagrams were quite simple. The idea is to construct as many words as possible, subject to the following two constraints:

1. Words must contain the middle letter/specific letter.
2. Words must have a minimum length of three letters and a maximum length of nine.

For the purposes of our project we will also add one additional rule:

3. Words must be contained in dictionary.

The multiplicative principle,⁶ is useful to determine how many possible puzzles can be constructed by choosing nine letters from the English alphabet. But this results in over counting.

Therefore we use a combinations functions since we consider ab and ba to have the same solutions. The correct number of possible 9 letter anagrams given a choice of all 26 letters is given by

$$nC_r = \frac{n!}{(n-r)!r!} = \frac{26!}{(26-9)!9!} = 3124550$$

This is the same combinations function we used to compute all the anagrams for the puzzles to be solved and to give us the data needed to begin solving them.

To be specific and allow the repetition of these results, using the itertools combinations module we gave python the 26 letter choices and told it to choose 9. We then iterated through it and wrote the results to a text file, the process is slow but the code was left simple because it is only needed once as all approximately 3.1 million anagrams puzzles are written to a text file to be saved.

Now we can talk about the method for solving the anagrams, with the given constraints.

First we made a function that would load the dictionary file we had and store it as a set for quick look-ups, included is a check if the length of the word is between 3 and 9 letters long. Since we had a letter constraint we built an index of words based on their letters, that way we can implement a binary search algorithm later for relatively faster computation. We then made a function to check if a word is valid, that is a word counts as a solution to the puzzle if all characters in the word found are included in the comparative word from the dictionary. These are included in a set to avoid duplicated results. If an anagram puzzle word contains the letter c for example its solutions, word from length 3 to 9 must all contain the letter c so the algorithm implemented would be to

1. Load and filter Dictionary
2. Load and sort anagrams
3. Read anagrams line by line
4. Use index to check if the letter is contained in both dictionary and anagram
5. Use binary search to run through sorted dictionary to find words faster
6. If contained write to file

Finally the results were generated using matplotlib and ran through the solutions file which contained the anagram and the number of words that anagram can make. It stored the number as a count and plots that against the number of anagrams, this is why we see a large number of anagrams with 0 words as a solution.

4 Conclusion

We find for some systems an exponential tail is inevitable but not for all, as the probability distribution has to have high initial event occurrence followed by growing decline in that probability.

This is common with events that follow the multiplicity process in determining the events probability. While many systems in nature and economy do follow this probability distribution, to generalize to all systems would be untrue without an exhaustive study on chaotic and non-chaotic systems.

It is however worth noting that the scope of the application of this probability distribution model is non-trivial and a large number of systems can be modeled to fit this correlation.

References

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